

Soc Internship day 17

Explain EDR vs XDR

Feature	EDR	XDR
Full Form	Endpoint Detection and Response	Extended Detection and Response
Focus	Endpoints only (PCs, laptops, servers)	Endpoints, network, email, cloud, identity
Data Sources	Endpoint logs	Multiple security sources
Visibility	Single device view	Organization-wide view
Threat Detection	Endpoint-based threats	Multi-stage attacks across systems
Investigation	Endpoint activities	Correlated attack chain
Response	Isolate endpoint, kill process	Block email, isolate device, block IP, disable account

EDR Example

A laptop is infected with ransomware.

EDR detects:

- Suspicious process: `ransomware.exe`
- File encryption activity
- Network connection to a malicious IP

EDR response:

- Kill process
- Isolate device
- Alert SOC team

XDR Example

An attacker sends a phishing email.

Attack chain:

1. Phishing email delivered
2. User clicks link
3. Malware downloaded
4. Malware connects to command-and-control server
5. Attacker logs in using stolen credentials

XDR detects:

- Email alert
- Endpoint malware alert
- Network connection alert

- Identity/login alert

XDR response:

- Quarantine email
- Isolate endpoint
- Block malicious IP
- Disable compromised account
- Create one incident containing the entire attack chain

Attack Scenario: Phishing Leading to Ransomware Infection

Title: Phishing Email Causes Ransomware Attack

Date/Time: 05-Jun-2026 10:15 AM

Attack Sequence**Step 1: Phishing Email**

- An employee receives an email claiming to be an invoice from a trusted vendor.
- The email contains a malicious attachment named `Invoice_June2026.docx`.

Step 2: User Interaction

- The employee opens the attachment.
- The document asks the user to enable macros.

Step 3: Malware Execution

- After macros are enabled, a malicious script downloads ransomware from an external server.
- The ransomware process starts running on the endpoint.

Step 4: Command and Control Communication

- The infected device connects to a malicious IP address:
 - IP: 185.23.11.5
- The attacker sends commands to the infected machine.

Step 5: File Encryption

- The ransomware encrypts files on the workstation and shared network drives.
- Users begin reporting inaccessible files.

Step 6: Detection

- EDR detects:
 - Suspicious PowerShell execution
 - Mass file modifications
 - Connection to a known malicious IP

Step 7: Response

- The SOC team isolates the endpoint.
- Malicious processes are terminated.

- The IP address is blocked on the firewall.
- Incident is escalated to L2/L3 analysts for further investigation.